

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech. (EC) Examination****November 2013****EC 204 Electronics Instruments & Measurement****Date: 28.11.2013, Thursday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define transducer. What are the differences between active transducer and passive transducer? [04]

(b) Give the balance conditions for ac bridge and dc bridge. [03]

Q - 2 (a) Draw and explain block diagram of Digital Storage Oscilloscope. [07]

(b) Explain Function Generator in detail with block diagram. [07]

OR

Q - 2 (a) Why a delay line used in the vertical section of Oscilloscope? Draw and explain delay line circuit. [07]

(b) Give the characteristics of Pulse. Explain pulse wave generator in detail. [07]

Q - 3 (a) Explain spectrum analyzer with figure. [05]

(b) Explain the basic circuit for measurement of frequency showing Gate Control F/F. [05]

(c) Derive the balance conditions for a basic bridge. [04]

OR

Q - 3 (a) Draw the structure of an LCD and explain its operation. [05]

(b) Explain the principle and operation of digital LCR meter with block diagram. [05]

(c) Discuss the limitations and errors of Wheatstone bridge. [04]

SECTION – II

- Q - 4 (a) Give the comparison between LED and LCD [04]
(b) What are the advantages of digital instruments over analog instruments? [03]
- Q - 5 (a) Describe the operation of Hay bridge and derive balance equation for resistance measurement. [05]
(b) A balanced AC bridge has following constants. [05]
Arm AB (Z_1): $R = 300 \text{ ohm}$ in series with $C = 0.265 \text{ uF}$ Arm BC (Z_2): $R = 450 \text{ ohm}$
Arm DA (Z_3): $R = 200 \text{ ohm}$ in series with $L = 15.9 \text{ mH}$ Arm CD (Z_4): unknown
The frequency of oscillation is 1 KHz . Find the constant of arm CD.
(c) Describe the classification of a transducers. [04]

OR

- Q - 5 (a) Which bridge is used for frequency measurement? Derive the frequency measurement equation for that bridge. [05]
(b) Write a short note on Piezo-electric transducer. [05]
(c) A Wheatstone bridge consists of following parameters. $R_1 = 10 \text{ k}\Omega$, $R_2 = 15 \text{ k}\Omega$ and $R_3 = 40 \text{ k}\Omega$. Find unknown resistance R_x [04]
- Q - 6 (a) Explain construction and operation of LVDT in detail. Also give advantages of LVDT [07]
(b) What is DAS? Explain with help of a block diagram. [07]

OR

- Q - 6 (a) What do you mean by strain gauge? Derive an expression for the gage factor in case of bonded wire strain gage. [06]
(b) Explain photo multiplier tube for photo electric transducer. [06]
(c) What is the limitation of Maxwell bridge? [02]